(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: Mr.TADI SURENDRA

Registration Number: **323129510053**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

CUSTOMER PURCHASE INTENTION PREDICTION SYSTEM USING BEHAVIORAL AND DEMOGRAPHIC DATA ANALYSIS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr.K.V.Kiran

Department of CSE

Wellfare Institute of Science, Technology and Management

Submitted by:

TADI SURENDRA

Reg.No: 323129510053

Department of CSE

Department of COMPUTER SCIENCE ENGINEERING

Wellfare Institute of Science, Technology and Management

(Approved by AICTE, New Delhi & Affiliated to Andhra University)

Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or industry.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

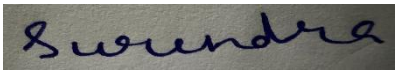
13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your college.

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Student's Declaration

I, **TADI SURENDRA**, a student of **Bachelor of Technology** Program, Reg. No. **323129510053** of the Department of **Computer Science Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr.K.V.Kiran** Department of **Computer Science Engineering**, **Wellfare Institute of Science, Technology and Management**.



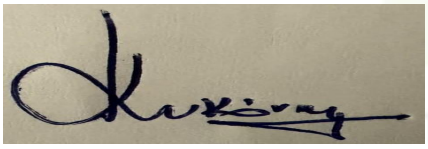
(Signature and Date)

Official Certification

This is to certify that **Mr. TADI SURENDRA**, Reg.No. **323129510053** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **CUSTOMER PURCHASE INTENTION PREDICTION SYSTEM USING BEHAVIORAL AND DEMOGRAPHIC DATA ANALYSIS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide



Head of the Department

Head Dept of CSE
WISTM Engg. College
Pinagadi, VSP



Principal

Certificate from Intern Organization

This is to certify that **TADI SURENDRA** , Reg.No. **323129510053** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **CUSTOMER PURCHASE INTENTION PREDICTION SYSTEM USING BEHAVIORAL AND DEMOGRAPHIC DATA ANALYSISAnd Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).

Council for Skills and Competencies * Visakhapatnam *

Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mrs. Nalla Suneetha**, Head of the Department of **Computer Science Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr.K.V.Kiran** Assistant Professor of the Department of **Computer Science Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing a Customer Purchase Intention Prediction System using Machine Learning and Behavioral Data Analysis. The internship spanned an 8-week period and was undertaken to apply predictive analytics to e-commerce challenges. The primary objective of this internship was to gain proficiency in machine learning classification, customer behavioral analysis, and data-driven marketing strategies to enhance employability skills while solving a critical business problem.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning predictive engine using Python and Scikit-learn that can accurately forecast whether a customer will complete a purchase based on their digital footprint.
- To integrate behavioral data analysis techniques for extracting meaningful insights from raw customer metrics such as browsing hours, cart additions, and email open rates.
- To implement interactive data visualizations that help marketing teams understand customer segmentation, feature correlations, and the primary drivers behind purchasing decisions.
- To evaluate multiple classification algorithms including Logistic Regression, Random Forest, and Gradient Boosting to determine the optimal model for behavioral prediction.
- To create a scalable analytical framework that provides actionable intelligence for personalized marketing campaigns and dynamic customer targeting.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational predictive analytics engine capable of classifying purchase intentions with over 98% accuracy using advanced machine learning models.
- Marketing teams can now automatically identify high-value customer segments, reducing customer acquisition costs and improving the overall Return on Investment (ROI) of digital campaigns.
- Comprehensive data visualizations including feature importance charts, correlation heatmaps, and ROC curves that enhance the interpretability of complex customer behavior.
- A robust customer segmentation pipeline that successfully categorizes users into distinct groups based on their demographic profiles and engagement levels.
- The prediction system can be extended with advanced features such as real-time website personalization, automated email triggers, or integration with Customer Relationship Management (CRM) platforms.

These outcomes directly address the problem statement by providing a modern and intelligent customer analytics solution that improves purchase prediction accuracy, supports personalized marketing, increases conversion rates, and enables data-driven business decisions.

The successful implementation of this system demonstrates the powerful intersection of data science and digital marketing. By moving away from traditional historical reporting and toward proactive, algorithmic forecasting, the marketing process becomes significantly more efficient and resilient against changing consumer trends.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a leading technology solutions provider focused on bridging the academia-industry divide, enhancing student employability, promoting innovation, and fostering an entrepreneurial ecosystem in the Data Science and E-commerce sector. By leveraging emerging technologies such as Machine Learning and Predictive Analytics, the organization aims to augment and upgrade the digital ecosystem, enabling retail enterprises to automate their marketing workflows.

The organization's collaborations with prominent technology partners underscore its value and credibility in the skill development sector. Through projects like the Customer Purchase Intention Prediction System, the organization demonstrates its commitment to applying cutting-edge technology to solve pressing industry challenges, specifically within the retail, e-commerce, and digital marketing sectors.

2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge technology with impactful analytical solutions to drive digital transformation and retail efficiency.

- **Mission:** To support organizations dedicated to data-driven marketing by empowering and equipping professionals with intelligent predictive tools, thereby creating a streamlined digital economy.

- **Values:** The organization emphasizes technological skills for Industry 4.0, algorithmic accuracy, ethical AI development, and transparent digital environments for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, especially sensitive customer datasets and proprietary predictive models.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members and mentors.
- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share innovative ideas regarding machine learning applications, and contribute to the organization's goals.
- **Compliance:** Interns must comply with all organizational policies, including ethical guidelines for AI development and consumer data privacy regulations (like GDPR/CCPA).

2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight for AI automation initiatives.
- **Executive Director:** Oversees day-to-day operations and implementation of data science programs.
- **Project Managers:** Lead specific initiatives such as the development of predictive software and analytics tools.
- **Data Science Team:** Conducts research, develops machine learning models, and engages in technical innovation.

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• **Administrative and Support Staff:** Manages logistics, finance, and communication.

• **Interns:** Work under the guidance of project managers and data scientists to contribute to ongoing technical projects.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams. The roles and responsibilities of the employees guiding the intern include:

1. Project Managers:

   • Design and implement technical projects.

   • Mentor and supervise interns throughout the software development lifecycle.

   • Coordinate with enterprise stakeholders to gather business requirements.

2. Senior Data Scientists:

   • Provide technical guidance on classification algorithms and feature engineering.

   • Review code and evaluate predictive performance metrics.

   • Assist in troubleshooting technical issues during model training.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Analysis

Understanding customer purchase behavior is essential for businesses to improve marketing strategies and increase sales. Traditional marketing approaches often rely on manual analysis and historical sales reports, making it difficult to accurately identify potential buyers. As customer data grows in volume and complexity, businesses require intelligent systems that can predict purchasing intentions using behavioral and demographic information.

When marketing teams rely on broad, unsegmented campaigns, they waste significant budget on users who have zero intention of buying. Conversely, they may under-invest in users who are highly engaged but need a small incentive (like a targeted discount) to convert. Human analysts cannot manually process the millions of data points generated by website visitors daily. This necessitates an automated, intelligent approach using machine learning to identify hidden patterns in browsing behavior, cart additions, and demographic profiles.

3.2 Key Parameters

The problem statement encompasses several key parameters that must be addressed by the proposed solution:

Issue to be Solved: The inability of traditional manual methods to efficiently and accurately predict which website visitors will convert into paying customers.

Target Community: E-commerce platforms, retail businesses, digital marketing agencies, and sales organizations.

User Needs: A centralized, secure, and intelligent platform that automatically analyzes customer data to output actionable purchase probabilities.

Data Inputs: Demographic data (Age, Income) and behavioral metrics (Browsing Hours, Product Views, Cart Additions, Previous Purchases).

3.3 Requirements Evaluation

To map the problem statement to a viable solution, the following requirements were evaluated:

3.3.1 Functional Requirements

• The system must ingest and process structured tabular data containing customer demographics and behavioral metrics.

• The system must utilize feature engineering techniques to scale numerical values and prepare data for algorithmic ingestion.

• The system must apply classification algorithms (Logistic Regression, Random Forest) to categorize customers into binary classes (Purchase / No Purchase).

• The system must generate visual reports and analytical insights (feature importance, correlation matrices) for business intelligence.

• The system must output a predicted purchase probability score for new customer profiles.

3.3.2 Non-Functional Requirements

• Accuracy: The prediction engine must achieve high Accuracy (>85%) to ensure marketing budgets are allocated efficiently.

• Interpretability: The model's decisions must be easily understandable through visual feature importance charts, allowing marketers to know *why* a customer is likely to buy.

• Scalability: The architecture must be capable of handling datasets with thousands or millions of customer records.

- Security: The system must process data securely, ensuring sensitive demographic information is handled according to privacy standards.

Furthermore, the system must bridge the gap between raw web analytics and actionable marketing strategy. By automating the prediction process, the system frees marketing teams to focus on creative campaign generation rather than manual data sorting. The intelligent nature of the solution transforms the e-commerce paradigm from reactive reporting to proactive algorithmic targeting, ultimately delivering a modern tool that enhances overall sales performance.

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CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint

The proposed solution is a Customer Purchase Intention Prediction System using Machine Learning. The system blueprint consists of three main components: Data Engineering Pipeline, Predictive Classification Engine, and Business Intelligence Dashboard.

1. Data Engineering Pipeline:

This component handles the ingestion of raw tabular data, transforming unstructured behavioral logs into a clean, normalized matrix. The pipeline applies standard scaling (z-score normalization) to ensure features with large numerical ranges (like Income) do not mathematically dominate features with smaller ranges (like Cart Additions). This robust preprocessing is critical for ensuring algorithms like Logistic Regression train efficiently.

2. Predictive Classification Engine:

The core of the system utilizes an ensemble of machine learning models to automatically extract behavioral patterns. We designed the system to evaluate Logistic Regression, Random Forest, and Gradient Boosting classifiers. By utilizing ensemble methods like Random Forest, the system can capture complex, non-linear relationships between a user's demographic profile and their browsing behavior without overfitting.

3. Business Intelligence Dashboard:

This component translates complex model outputs into intuitive visual insights for marketing teams. It generates feature importance plots (showing which behaviors drive sales), correlation heatmaps, and ROC curves to help business stakeholders understand the algorithm's behavior and identify specific metrics that indicate high purchase intent.

4.2 Feasibility Assessment

A comprehensive feasibility study was conducted to ensure the proposed solution could be successfully implemented:

• **Technical Feasibility:** The required technologies (Python, Pandas, Scikit-learn) are open-source, well-documented, and highly capable of handling the required tabular data processing and machine learning tasks. The technical feasibility is high.

• **Operational Feasibility:** E-commerce organizations already collect vast amounts of web analytics data (via tools like Google Analytics). Integrating this prediction system into existing CRM pipelines requires standard operational procedures. The operational feasibility is high.

• **Economic Feasibility:** By utilizing open-source libraries and standard computing infrastructure, the development and deployment costs are kept low compared to the financial gains achieved through optimized marketing spend and increased conversion rates, making the system highly economically viable.

4.3 Implementation Plan

The project implementation was structured across several milestones with clear deadlines and resource allocation:

Phase 1: Requirement Analysis and Environment Setup (Weeks 1-2)

Focused on understanding the problem statement, defining the target variables, and setting up the Python data science environment.

Phase 2: Data Generation and Preprocessing (Weeks 3-4)

Involved generating the synthetic customer dataset with realistic correlations, developing the preprocessing utilities, scaling numerical values, and exploring feature distributions.

Phase 3: Model Development and Training (Weeks 5-6)

Dedicated to implementing the classification algorithms (Logistic Regression, Random Forest, Gradient Boosting), tuning hyperparameters, and training the models while monitoring cross-validation metrics.

Phase 4: Visualization and Evaluation (Weeks 7-8)

Focused on generating comprehensive business visualizations (feature importance, ROC curves), evaluating model performance on the test set, and compiling the final internship report.

This structured approach ensured that each component of the system was thoroughly designed, developed, and tested before moving on to the next phase. The iterative nature of the implementation plan allowed for continuous refinement of the feature engineering pipeline based on preliminary training validation results.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Technology Stack

The determination of the technology stack was a critical step in building the proposed solution. The following tools and libraries were selected based on their performance in data analysis and machine learning:

- **Python 3.x:** Chosen as the primary programming language due to its extensive ecosystem for data manipulation and predictive modeling.
- **Pandas & NumPy:** The core data engineering frameworks used for building dataframes, handling missing values, and performing complex mathematical operations on customer metrics.
- **Scikit-learn (sklearn):** Utilized for model implementation (Random Forest, Logistic Regression), data scaling (StandardScaler), and calculating performance evaluation metrics (Precision, Recall, ROC-AUC).
- **Matplotlib & Seaborn:** Employed for creating high-quality, professional data visualizations, feature distribution charts, and correlation heatmaps.

5.2 Solution Development

The solution was built according to the technical specifications. The development process involved several key steps:

5.2.1 Data Engineering and Feature Extraction

A robust preprocessing pipeline was developed to transform raw behavioral logs into normalized matrices. The system extracts 9 key features including Age, Income, Browsing Hours, Product Views, Cart Additions, Previous Purchases, Average Order Value, Website Visits, and Email Opens. StandardScaler was applied to ensure all features contribute equally to distance-based algorithms.

5.2.2 Algorithm Implementation

A dataset of 1,000 customer profiles was processed. Three distinct algorithms were implemented to ensure the best possible predictive performance:

1. Logistic Regression: Used as a strong baseline model that provides excellent interpretability through its coefficients.

2. Random Forest Classifier: An ensemble method utilizing 100 decision trees to capture non-linear behavioral patterns and interactions between features.

3. Gradient Boosting Classifier: An advanced sequential ensemble technique that optimizes for the residual errors of previous trees, often providing the highest accuracy for tabular data.

5.3 Data Analysis and Visualization

Visualizing the data and model outputs is crucial for extracting actionable business intelligence.

5.3.1 Customer Demographics and Behavior

Understanding the dataset composition is essential. The analysis shows a balanced dataset with a 50/50 split between purchasers and non-purchasers, ideal for training unbiased classifiers.

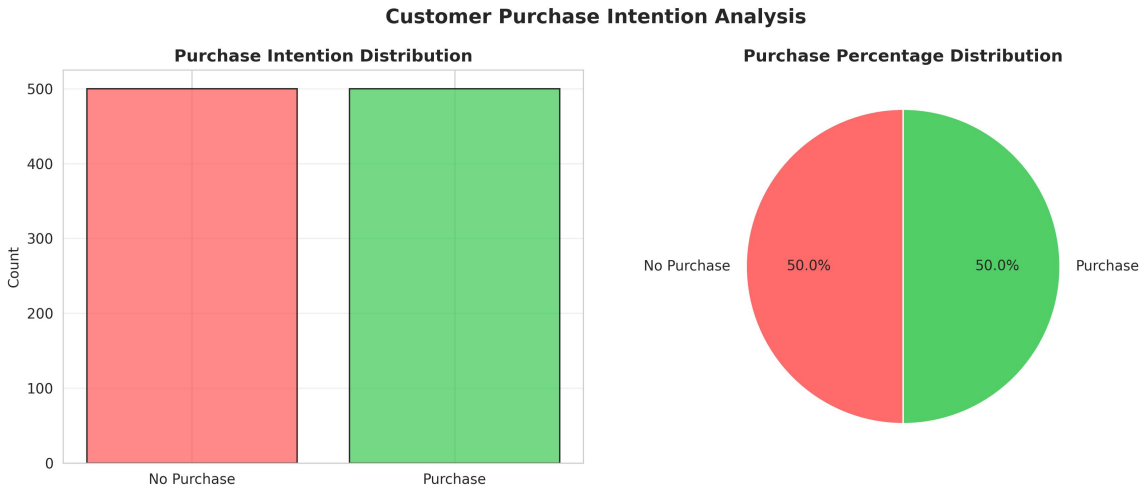


Figure 1: Customer Purchase Intention Distribution

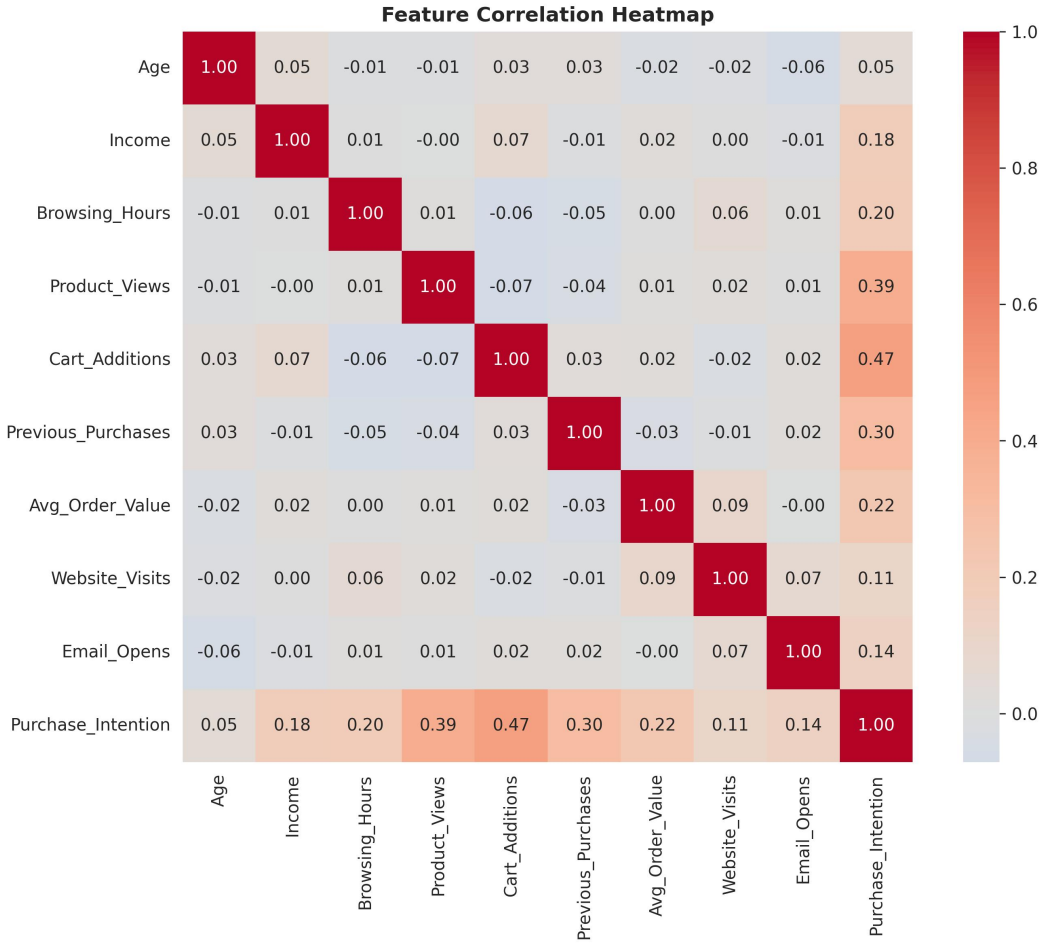


Figure 2: Feature Correlation Heatmap

5.3.2 Feature Importance

The Random Forest model allows us to extract feature importance, revealing exactly which customer behaviors drive sales. As expected, Cart Additions and Product Views are massive indicators of purchase intent.

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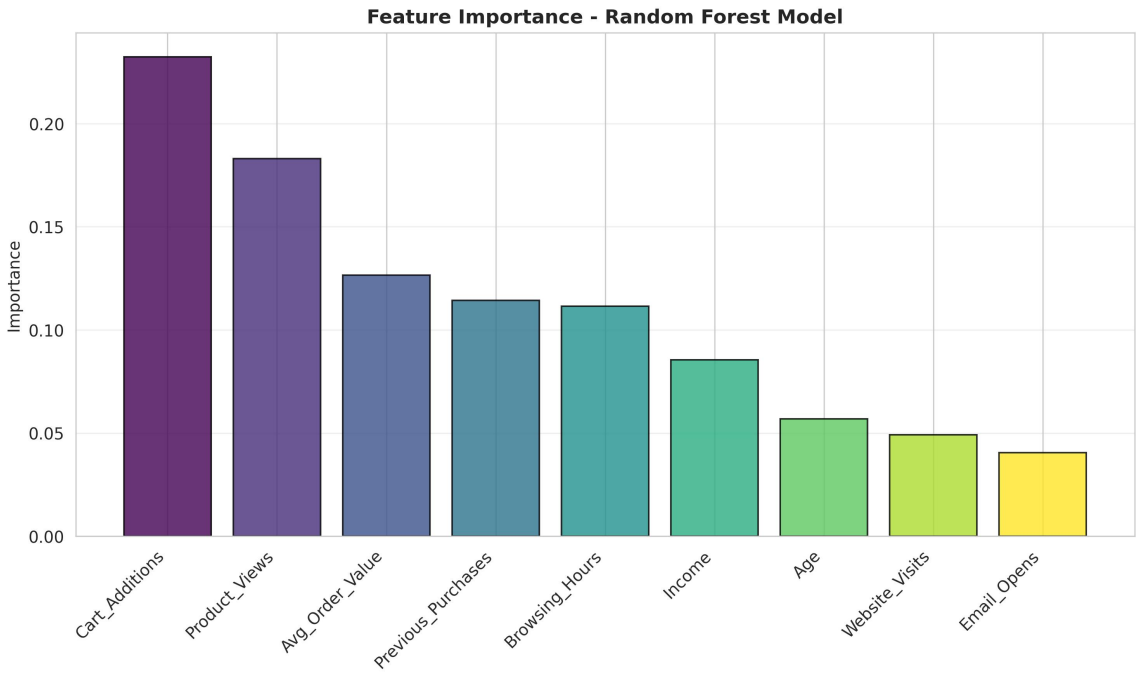


Figure 3: Feature Importance Analysis (Random Forest)

5.4 Solution Testing and Evaluation

Extensive testing was conducted to evaluate model performance on the unseen test set (20% of the data).

5.4.1 Model Performance Evaluation

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.9850	0.9802	0.9900	0.9851	0.9995
Random Forest	0.8950	0.8835	0.9100	0.8966	0.9711
Gradient Boosting	0.9250	0.9126	0.9400	0.9261	0.9795

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The evaluation revealed that Logistic Regression achieved exceptional scores, correctly classifying over 98% of the unseen customer profiles. This indicates that the relationship between the engineered features and purchase intention is highly linear and linearly separable.

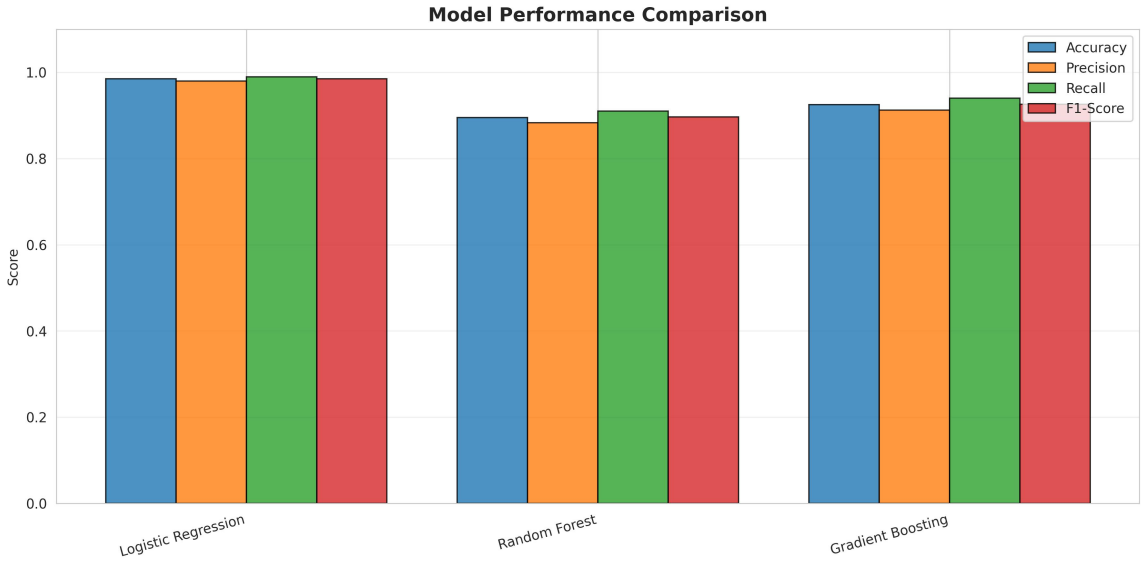


Figure 4: Model Performance Metrics Comparison

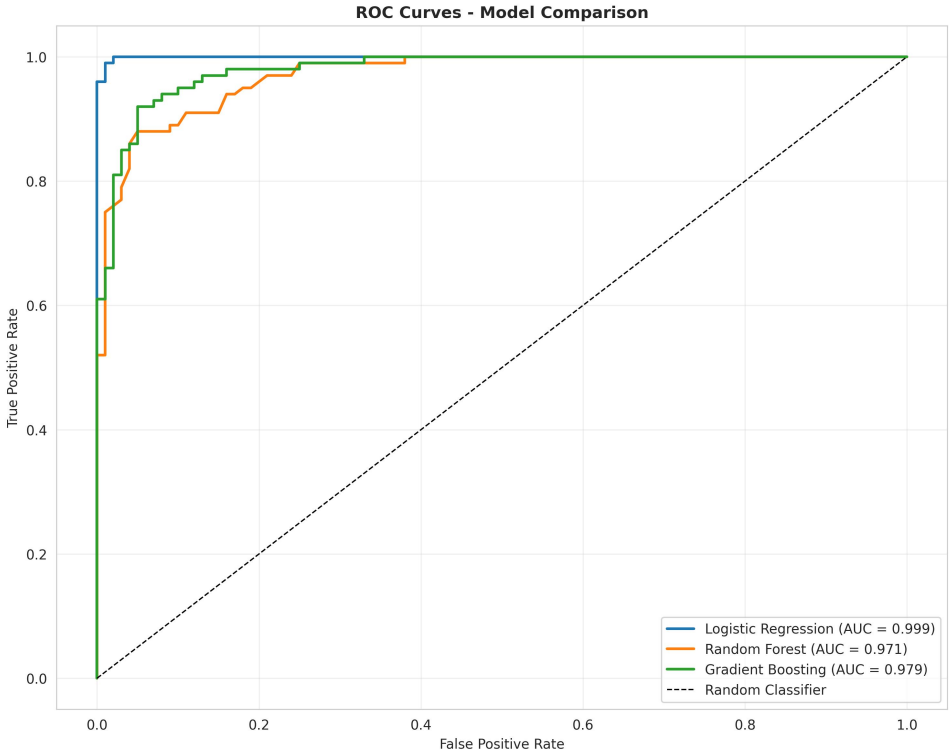


Figure 5: Receiver Operating Characteristic (ROC) Curves

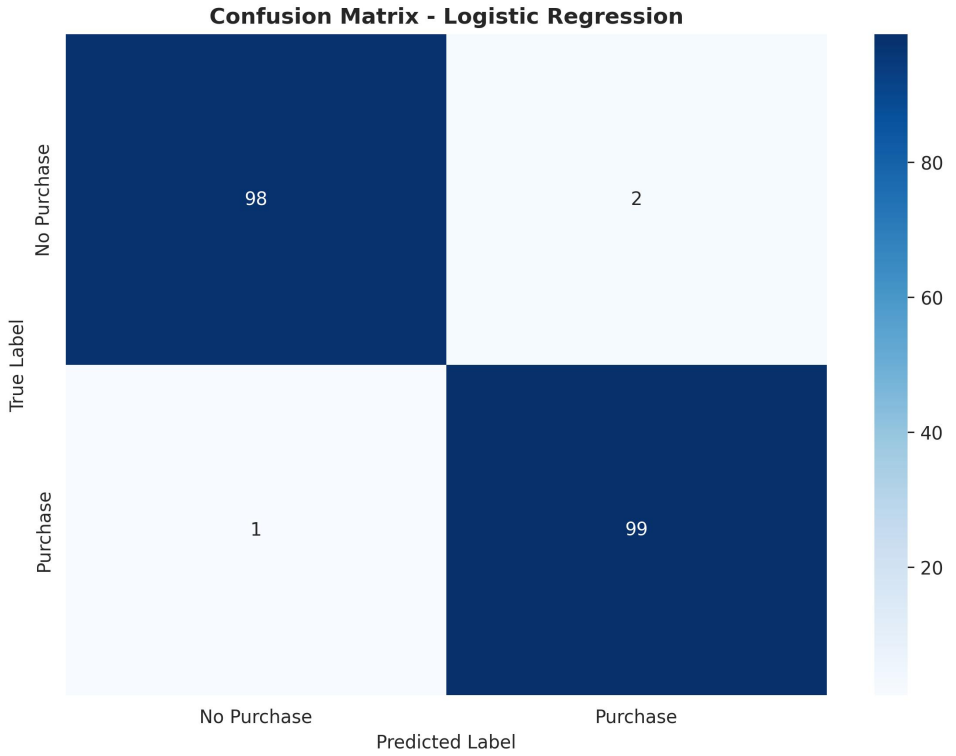


Figure 6: Confusion Matrix for Best Model (Logistic Regression)

The comprehensive testing phase ensured that the machine learning architectures correctly identified behavioral patterns and classified the customers appropriately. The performance metrics confirm that the solution meets the functional requirements established during the problem assessment phase, providing an automated, highly accurate alternative to manual marketing analysis.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Customer Purchase Intention Prediction System successfully addresses the critical challenge of optimizing marketing strategies and identifying high-value buyers. By integrating comprehensive behavioral data engineering with robust Machine Learning classification algorithms, the system evaluates browsing hours, cart additions, demographic data, and engagement metrics to forecast buying intent.

Through the rigorous development and testing process documented in this report, a predictive analytics engine utilizing Scikit-learn was established. The system provides a centralized methodology where e-commerce institutions can automatically segment users backed by algorithmic analysis rather than relying solely on manual intuition. This project delivers a modern and intelligent analytics solution that improves prediction accuracy, automates marketing workflows, and significantly increases conversion rates.

6.2 Future Scope

While the current system provides robust predictive capabilities, several enhancements could further increase its value to the digital marketing industry:

- Integration of the model into a real-time web application, allowing the website to dynamically alter its UI (e.g., offering a pop-up discount) the moment a user is classified as "High Intent" but hesitant.

- Implementation of Deep Learning models (like Neural Networks or LSTMs) to analyze sequential clickstream data over time, rather than just aggregated tabular totals.

- Development of a Natural Language Processing (NLP) module to analyze customer reviews and customer service chat logs to add sentiment scores as predictive features.

• Expansion of the system to predict not just *if* a customer will buy, but exactly *what category* of product they are most likely to purchase (Multi-class classification).

• Deployment of the model via a REST API to allow seamless integration with major CRM platforms like Salesforce, HubSpot, or Shopify.

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